GitHub Repository: https://github.com/hafsaahmed05/CS449_Solitaire

Sprint #0 Report

Instructions

Objectives

- Make decisions on the Solitaire software development project.
- Learn unit testing and GUI programming in the language of your choice.

Deliverables and Grading Policy

Read the "CS 449 Homework Overview" document **carefully** and make the key decisions for the software development. Use the following template to complete your report.

1. Key Decisions of the Solitaire Project (2 points)

Object-oriented programming language (if not using OOP, then justify the use of the language)	Java
GUI library (strongly encouraged)	JavaFX
IDE (Integrated Development Environment)	VSCode or Eclipse (I couldn't get Java on VScode working by the time Sprint0 was due, but I would prefer working in VSCode. I will do my best to figure it out by the next sprint. I did Sprint0 in Eclipse though.)
xUnit framework (e.g., JUnit for Java)	JUnit
Programming style guide (must read it carefully)	https://google.github.io/styleguide/javaguide.html
Project hosting site	Github.com
Other decisions if applicable	

Sample programming style guides:

- Google Java Style Guide: https://google.github.io/styleguide/javaguide.html
- Google C++ Style Guide: https://google.github.io/styleguide/cppguide.html
- Google Python Style Guide: https://google.github.io/styleguide/pyguide.html

2. Unit testing (4 points)

Find a tutorial on the unit test framework you have chosen and write at least two xUnit tests of a program you have written or found elsewhere. Attach here (1) the screenshot of your program execution and (2) the source code of your program.

For example, Junit5 user guide: https://docs.junit.org/5.10.3/user-guide/index.html

MinMax.java

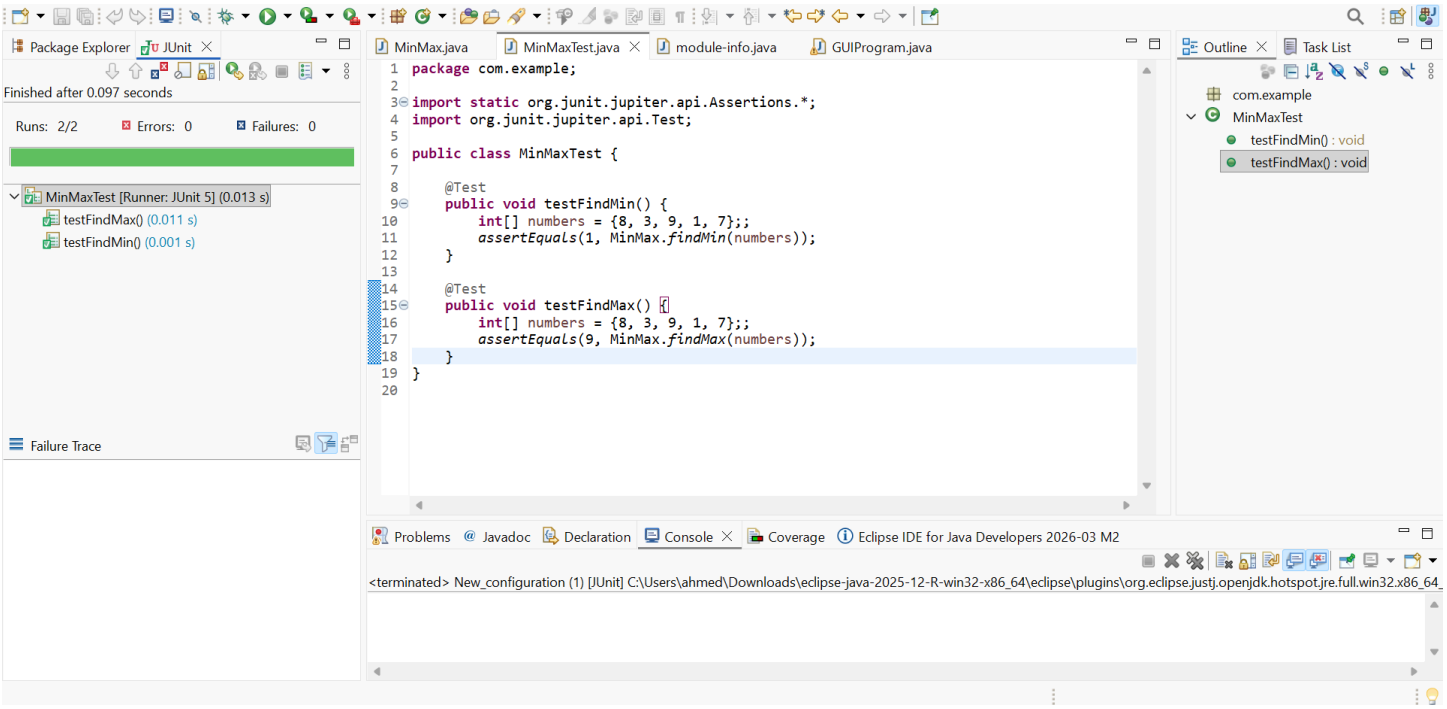
```
package com.example;

public class MinMax {
```

```java
    // Returns the minimum value in an array
    public static int findMin(int[] numbers) {
        int min = numbers[0];
        for (int n : numbers) {
            if (n < min) min = n;
        }
        return min;
    }

    // Returns the maximum value in an array
    public static int findMax(int[] numbers) {
        int max = numbers[0];
        for (int n : numbers) {
            if (n > max) max = n;
        }
        return max;
    }
}
```

MinMaxTest.java

```java
package com.example;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

public class MinMaxTest {

    @Test
    public void testFindMin() {
        int[] numbers = {8, 3, 9, 1, 7};
        assertEquals(1, MinMax.findMin(numbers));
    }

    @Test
    public void testFindMax() {
        int[] numbers = {8, 3, 9, 1, 7};
        assertEquals(9, MinMax.findMax(numbers));
    }
```

Output:



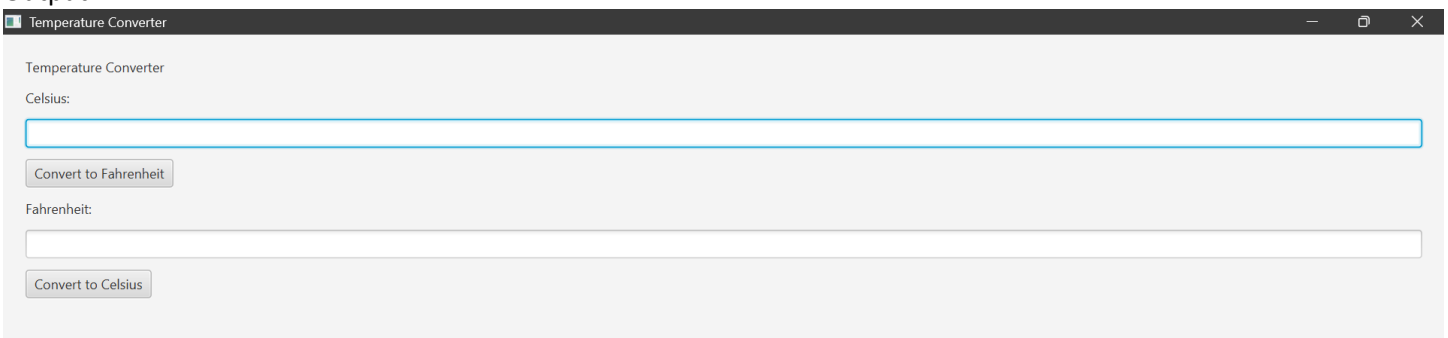
3. GUI programming (4 points)

Write a GUI program in the language you have chosen for your Solitaire project. The GUI of your program must include text, lines, a check box, and radio buttons. In this assignment, any GUI program of your own work is acceptable.

Attach here (1) the screenshot of your program execution and (2) the source code of your program.
Source Code:

```
package com.example;

import javafx.application.Application;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.TextField;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class GUIProgram extends Application {

    @Override
    public void start(Stage primaryStage) {
        // Create labels
        Label titleLabel = new Label("Temperature Converter");
```

```
        Label celsiusLabel = new Label("Celsius:");
        Label fahrenheitLabel = new Label("Fahrenheit:");
        Label resultLabel = new Label("");

        // Create text fields
        TextField celsiusField = new TextField();
        TextField fahrenheitField = new TextField();

        // Create buttons
        Button convertToFButton = new Button("Convert to Fahrenheit");
        Button convertToCButton = new Button("Convert to Celsius");

        // Set button actions
        convertToFButton.setOnAction(e -> {
            try {
                String celsiusText = celsiusField.getText();
                double celsius = Double.parseDouble(celsiusText);
                double fahrenheit = (celsius * 9.0/5.0) + 32;
                fahrenheitField.setText(String.format("%.2f", fahrenheit));
                resultLabel.setText(celsius + "°C = " + String.format("%.2f", fahrenheit) +
"°F");
            } catch (NumberFormatException ex) {
                resultLabel.setText("Please enter a valid number!");
            }
        });

        convertToCButton.setOnAction(e -> {
            try {
                String fahrenheitText = fahrenheitField.getText();
                double fahrenheit = Double.parseDouble(fahrenheitText);
                double celsius = (fahrenheit - 32) * 5.0/9.0;
                celsiusField.setText(String.format("%.2f", celsius));
                resultLabel.setText(fahrenheit + "°F = " + String.format("%.2f", celsius) +
"°C");
            } catch (NumberFormatException ex) {
                resultLabel.setText("Please enter a valid number!");
            }
        });

        // Create layout
        VBox layout = new VBox(10);
        layout.setPadding(new Insets(20, 20, 20, 20));
        layout.getChildren().addAll(
            titleLabel,
            celsiusLabel,
```

```
            celsiusField,
            convertToFButton,
            fahrenheitLabel,
            fahrenheitField,
            convertToCButton,
            resultLabel
        );

        // Create scene and show stage
        Scene scene = new Scene(layout, 300, 350);
        primaryStage.setTitle("Temperature Converter");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Output: